

Accumulator EE

Katherine Chen

BMS BMS BMS BMS BMS

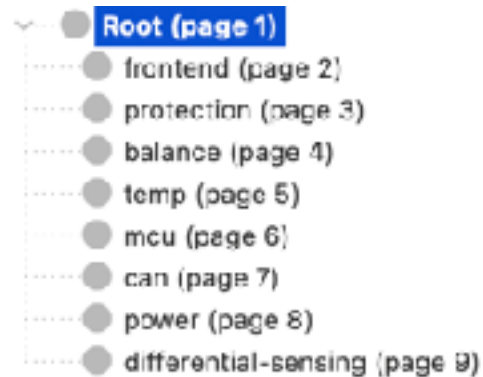


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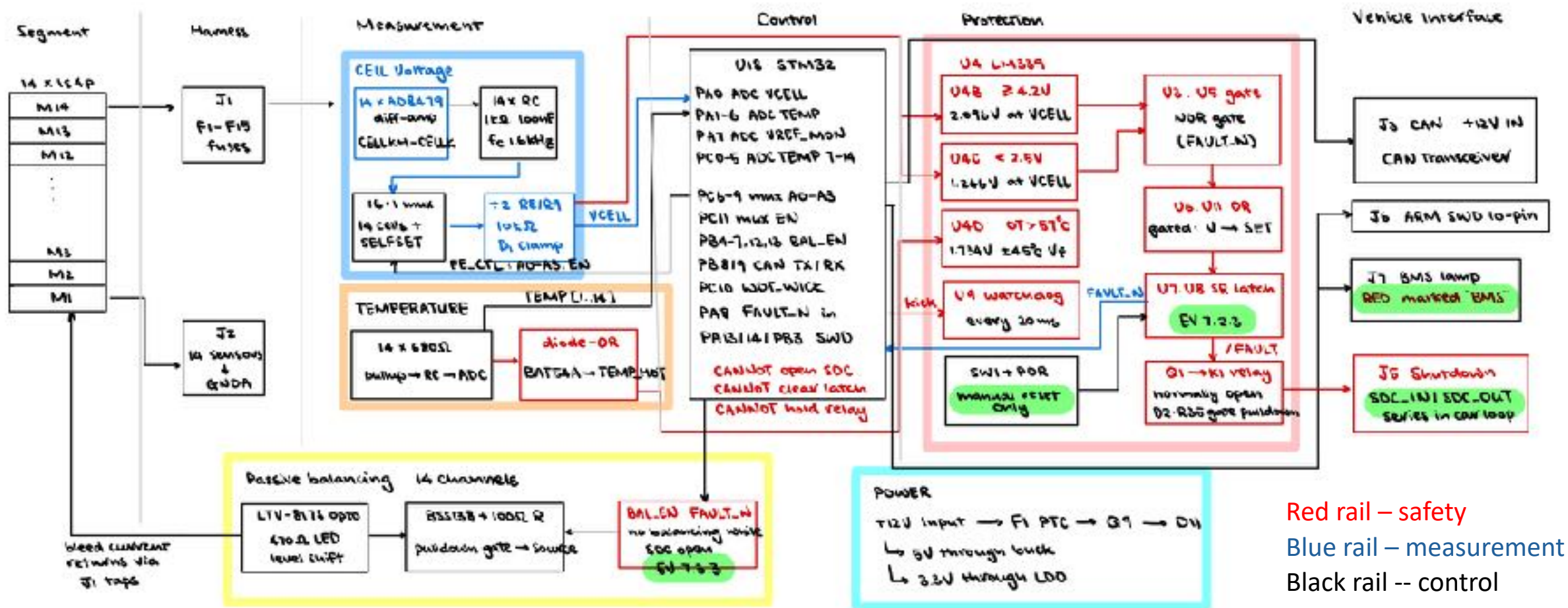
BMS for one 14s4p segment: Enepaq 1x4p modules
35–58.8 V ~10 Ah

Every cell needs to be measured bc:
The pack voltage hides the worst cell.
50 V = 30 healthy cells + 1 at 2.1 V

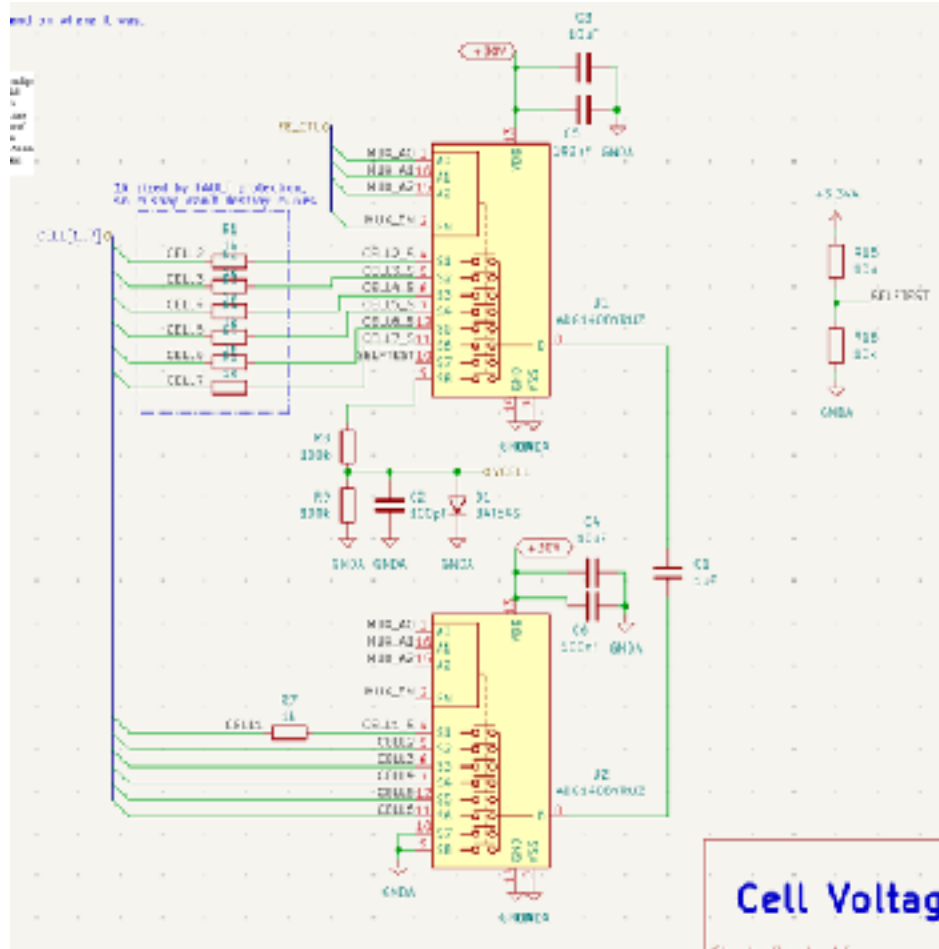
Reference FSAE_Rules_2026_V1 regulations



Block Diagram



Cell Voltage – Initial solution



This is for 6S modules!

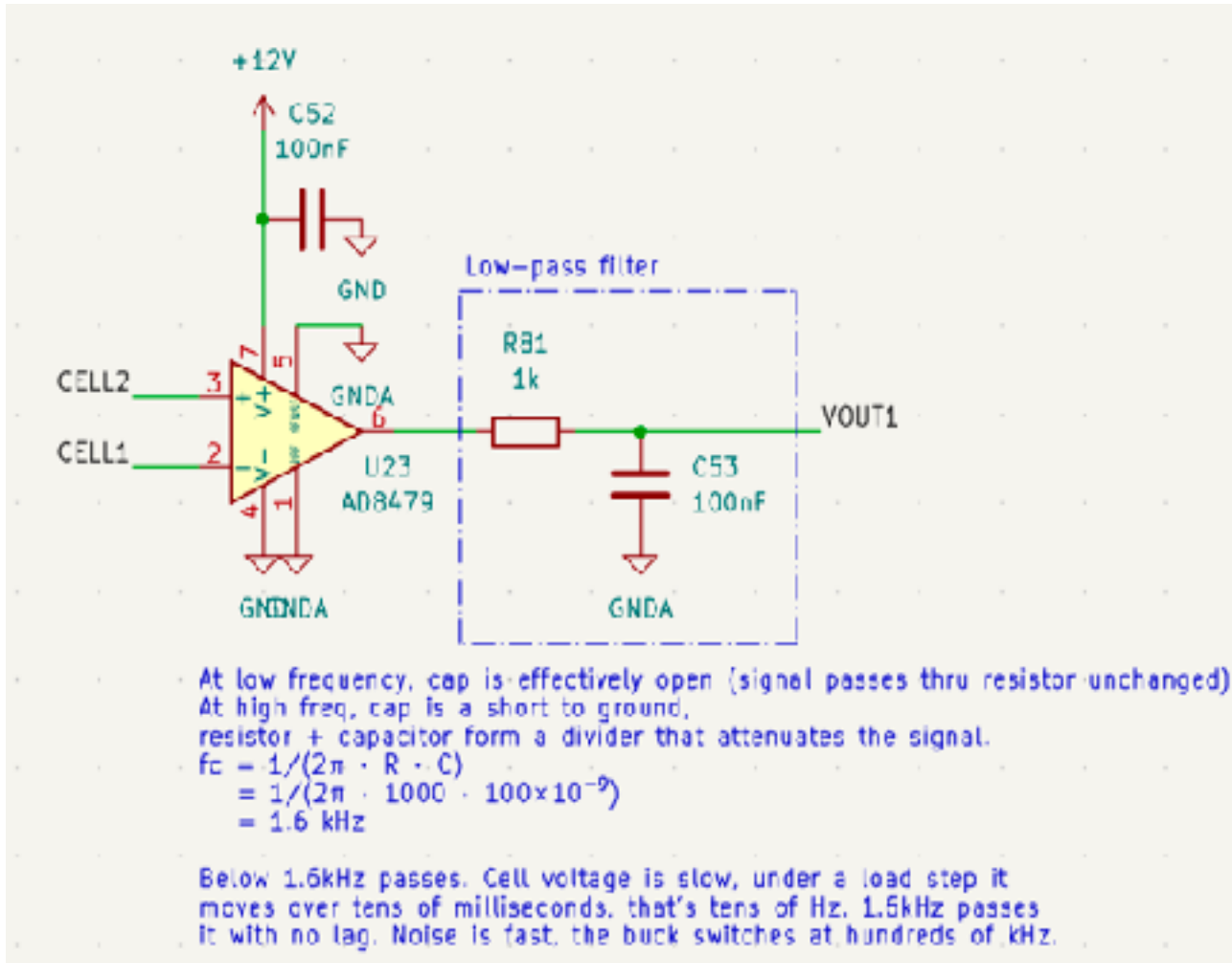
The idea: Flying Capacitor

- Cell 14's terminals sit at 50–54 V; ADC reads 0–3.3 V against its own ground
- Muxes connect a capacitor across one cell → disconnect both ends → connect it to ground and the ADC. It holds the cell voltage with no memory of where it was
- One shared path → one gain error, common to all cells.

Problem:

- We can NOT get all 14 series cells at once → sequential
- Capacitor starts at different capacitance everytime

Cell Voltage – optimized version ...



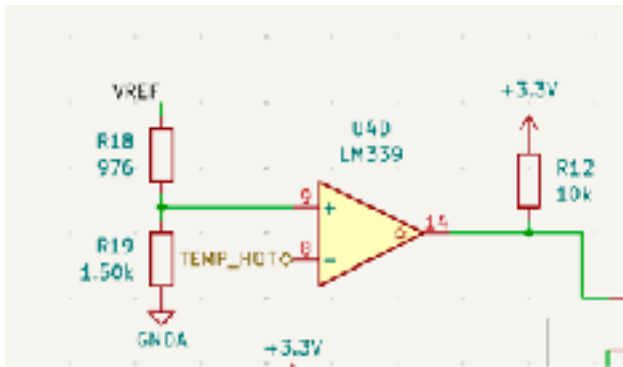
VOUT goes to ADC pins on MCU

The idea: Differential Sensing

- One difference amplifier per cell
→ measure all cells at same time
- inputs straight on the two terminals
Output = (CELL k+1) – (CELL k)
- Added a low-pass filter for noise filtering
→ output stable signal

Temperature Sensing

Effectively, ADC can read temperature data precisely and compare to 60C (with a comparator U4!).

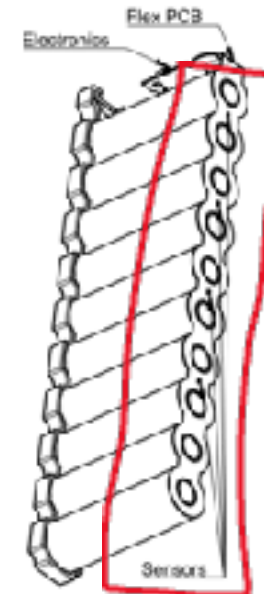


If firmware is dead, watchdog catches that.

But what if code is running but wrong, e.g. reads the wrong channel, has a stale value...

OR how can we know temp data while coding isn't running

From module datasheet

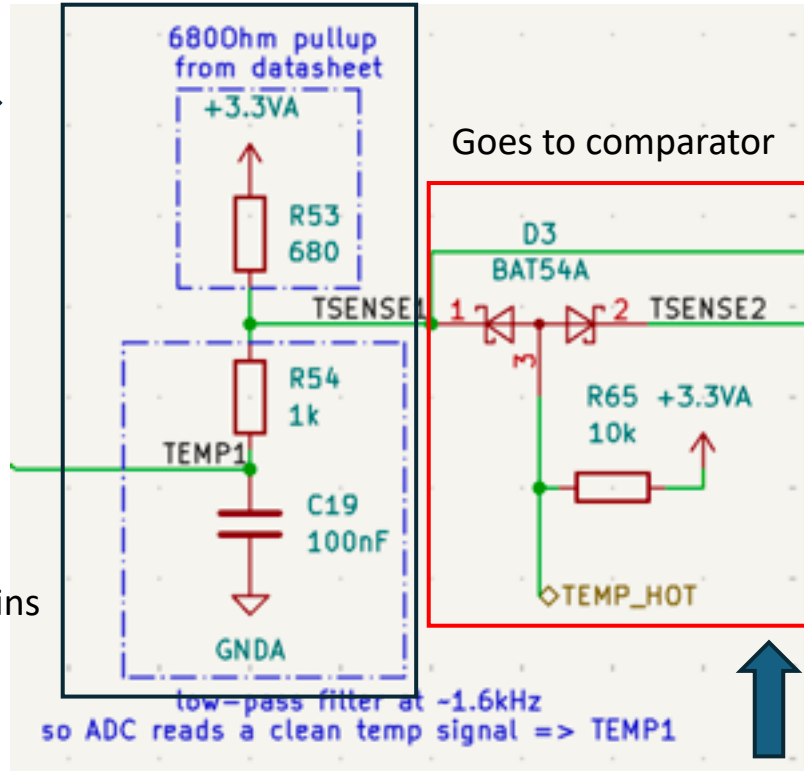


FEATURES

- Small size: 237-411 Wh per liter
- Low weight: 183-193 Wh per kg
- Individually fuse-protected cells
- Ultra low and equal self-discharge
- Rapid prototyping of battery pack
- Convenient thermal control
- Built-in temperature sensors
- UL94-V0 rated, fire-retardant plastics
- UN38.3 certified ¹⁴

The sensor is a specially-made temperature-variable voltage shunt reference. In simple words, it acts as a zener diode, whose voltage drop depends on temperature. It requires a pull-up resistor (680Ω) to operate at cell voltage level. For convenience, the module can be used to power the sensor as given in test circuit in Figure 4.

Goes to ADC pins



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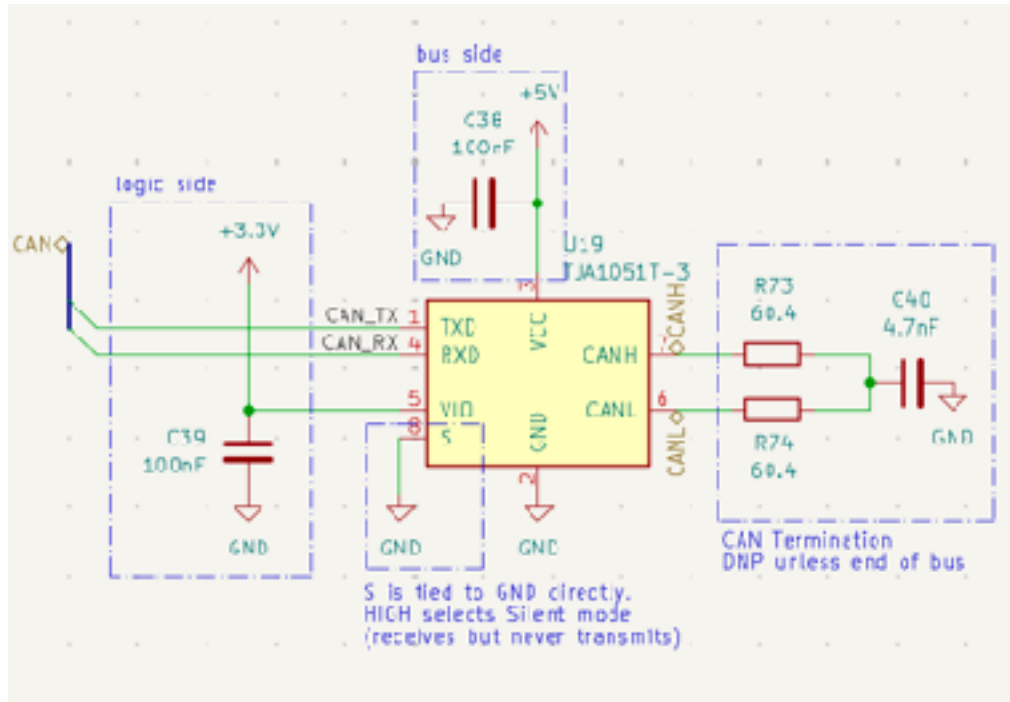
- diode is a one way valve, current can flow from anode to cathode if $V_{anode} \geq V_{cathode}$

- TEMP_HOT at 1.V: both diodes shut
- TEMP_HOT at 1.7V: diode2 opens, current flows from TEMP_HOT through D2 to sensor 2

That current holds TEMP_HOT at 1.7V bc D2 drains it as fast as the resistor supplies it. D1 is never opened.

Net	From	To	Purpose
TENP1 ... TENP14	each sensor, filtered	MCU ADC pins	firmware knows every module
TENP_HCT	all sensors, via diodes	U4D on the protection sheet	hardware trips on the hottest

CAN transceiver + termination (DNP)



CAN transceiver: a translator between 3.3 V logic on MCU and differential signal on CAN harness.

CAN differential signaling

- No master and slave, every address wants to talk at same time
- Opposite polarity → noise lands on both and cancels at the receiver
- A node transmits when the bus is idle; collisions resolve by address; lower address has higher priority <== CAN actively low (prefer 0 over 1)
- e.g cell temp data from BMS to datalogger, steering info from dashboard to control movements

How BMS interface with the car? CAN!

- BMS is a node in the overall CAN line, usually not termination node → DNP
- (no starting node and terminating node → if CAN line left floating, signals will reflect back → put two resistors on each end)

BMS interface (other than CAN)

- Shutdown circuitry (BMS fault) : relay contactors in series with the car's loop, SDC_IN/SDC_OUT on J5
 - “The BMS, IMD, and BSPD parts of the Shutdown Circuit must be **Normally Open**” (EV.7.1.3)
- BMS indicators (LEDs) for drivers

EV.7.3.5 If the BMS detects one or more of the conditions of EV.7.3.4 above, the BMS must:

- Open the Shutdown Circuit EV.7.2.2
- Turn on the BMS Indicator Light and the Traction System Status Indicator EV.5.11.5
The two lights must stay on until the BMS is manually reset EV.7.2.3

EV.7.3.6 The BMS Indicator Light must be:

- Color: Red
- Clearly visible to the seated driver in bright sunlight
- Clearly marked with the lettering “BMS”

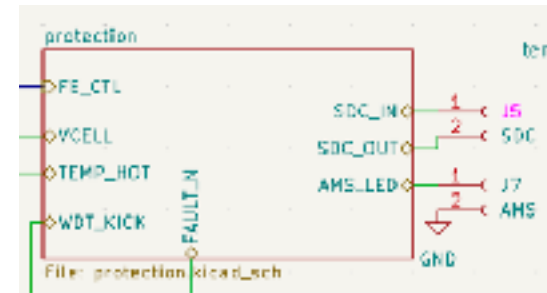
- RMS fault limits:

EV.7.5.2 Temperatures (considering measurement accuracy) must stay below the lower of the two:

- The maximum cell temperature limit stated in the cell data sheet
- 60°C

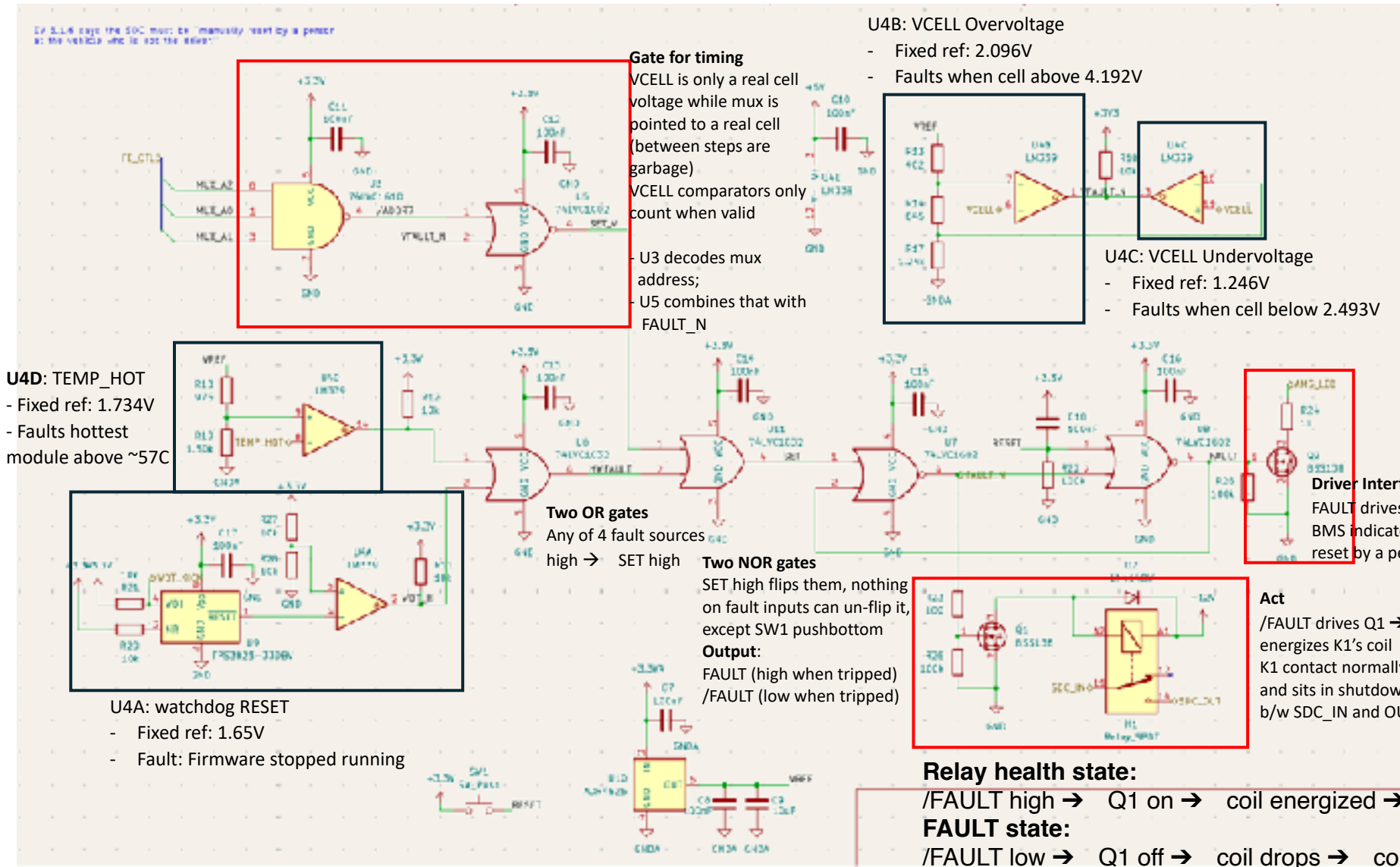
EV.7.4.2 Cell Voltage levels must stay inside the permitted minimum and maximum cell voltage levels stated in the cell data sheet. Measurement accuracy must be considered.

(reference: SONY/Murata VTC6 datasheet & FSAE_Rules_2026_V1)



Battery voltage (V)			Battery capacity	
Min.	Typ.	Max.	Ah	Wh
2.5	3.6	4.2	5	10.5
			8	21.6
			9	23.4
			12	30.2
			15	39.6
			18	47.4
			21	55.8
			24	64.8
			27	73.2
			30	81.0
			36	97.2
			42	113.4
			48	129.6
			54	145.8
			60	162.0

How system monitors unsafe conditions → BMS fault



When a cell or a module has left its safe range, open the shutdown circuit.

U4: LM339 (four comparators)

Each one holds a fixed voltage on one input and a live signal on the other, and outputs "yes/no"

- Fixed voltage comes from resistor dividers off 2.5V ref
- R13/R14/R17: cell
- R18/R19: temperature

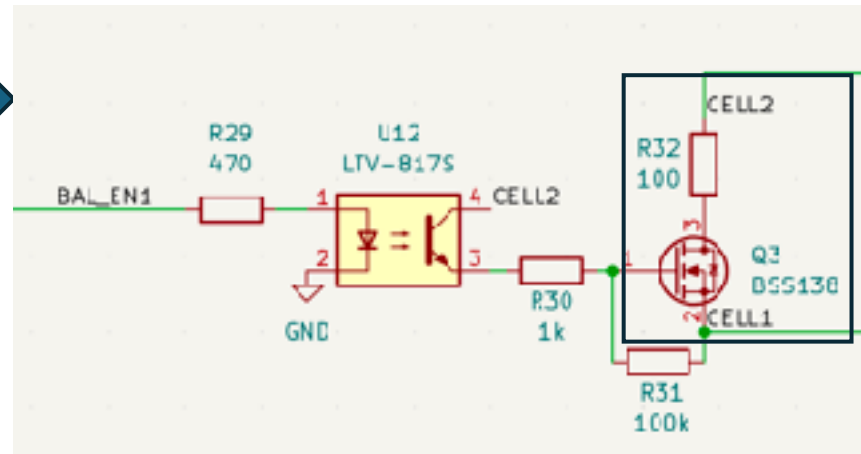
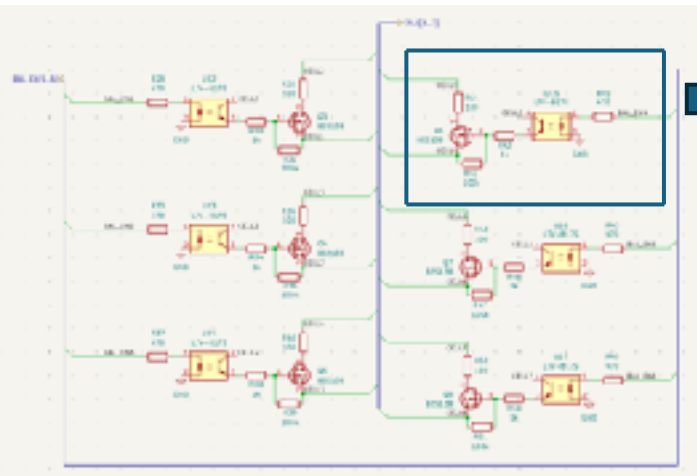
- LM339's output are open-collector: only pull down, each need pull-up resistor

Cell Balancing

Why balancing: Each cell has a slightly different capacity, each self-discharges at a slightly different rate, and the ones in the middle of the pack run warmer and age faster. Thus, the cells drift apart.

That's a problem because the pack is limited by its extremes.

- On charge, the **fullest** cell hits 4.2 V first and the whole pack must stop, the others aren't full yet.
 - On discharge, the **emptiest** cell hits 2.5 V first and the pack must stop, the others still have charge.
- Balancing pulls the cells back together so the pack can use its whole capacity



Cell 6's switch has its source at 18.5 V. If the MCU drives that gate with a 3.3 V signal referenced to ground, the gate is 15 V *below* the source. The transistor doesn't turn on.



Passive balancing:

- Close switch: cell drains through resistor
- Open switch: comes down to match others

Firmware decides which switch to close based on voltage data from frontend:
bleed highest, stop when they match w lowest

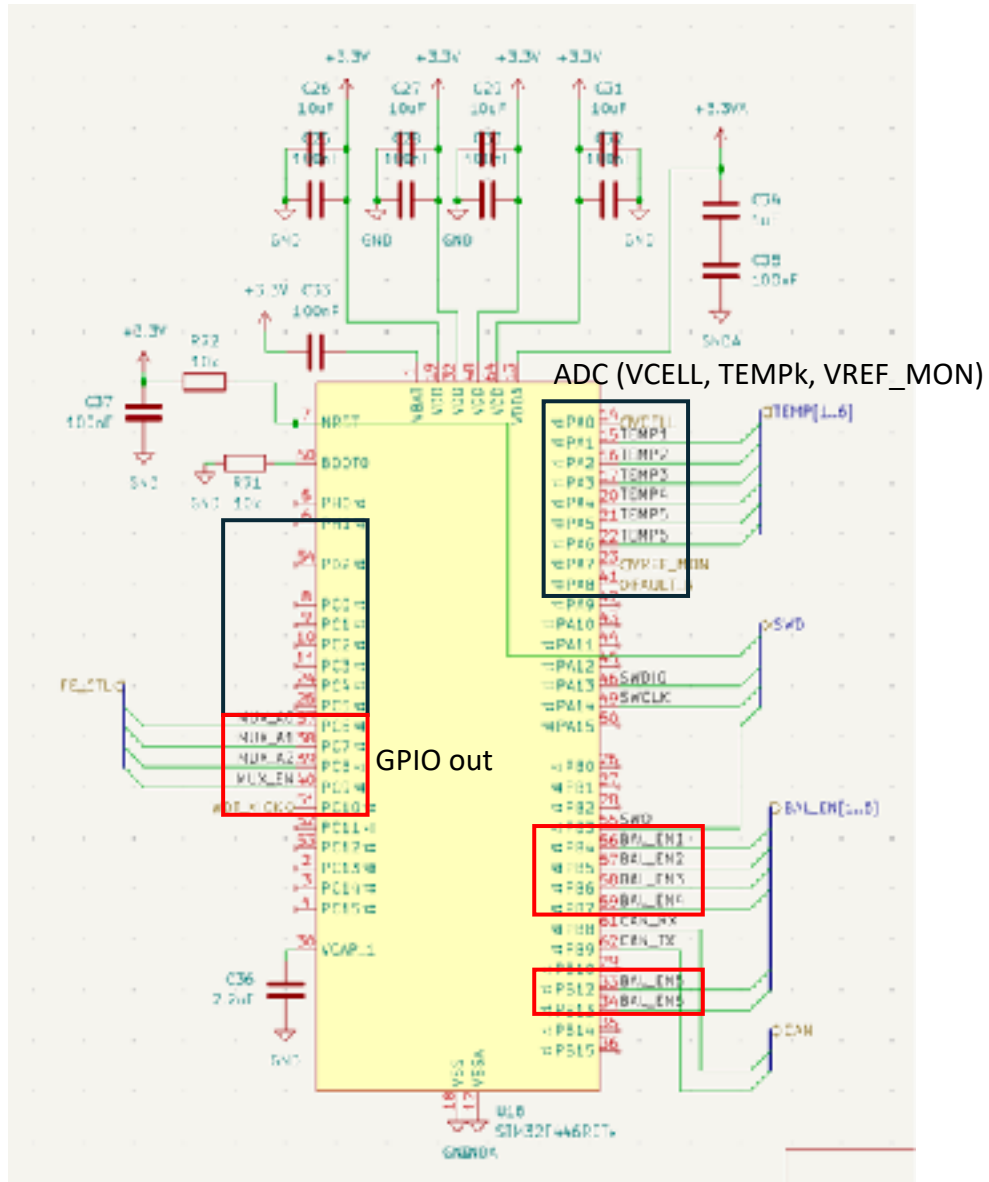
Optocoupler: MCU lights the LED

The light turns on the phototransistor, which is wired between the cell's positive terminal and the FET's gate, so it pulls the gate up relative to the source regardless of where that cell sits

Bonus:

this is galvanic isolation for free, which is what the rules want between the tractive side and the logic side anyway.

MCU pinout



STM32F446RE

- **ADC:** PA0 VCELL, PA1–6 TEMP, PA7 VREF_MON, PC0–5 spare (14S)
- **GPIO out:** PC6–9 mux, PB4–7, 12, 13 BAL_EN, PC10 WDT_KICK
- **GPIO in** PA8 FAULT_N
- **CAN1:** PB8 RX, PB9 TX
- **SWD:** PA13/PA14/PB3

→ ADC ref = VDDA, corrected from the 2.500 V on PA7

→ WDT_KICK is a GPIO in the main loop

→ 14S: 16 of 16 ADC channels

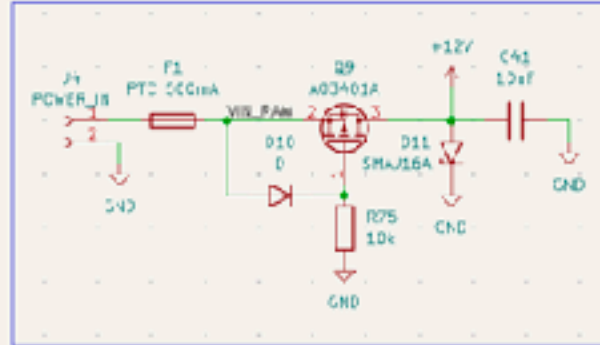
Power

Battery pack only powers motors and invertors.

Everything on BMS is powered from 12V LV system.

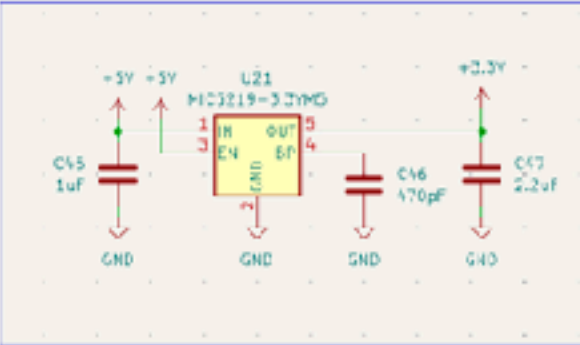
- PTC fuse + P-FET reverse-polarity blocker for input protection
- Two stage to 3.3V (Buck + LDO) for efficiency + tripping switching noise
- FB1 for analog parts

Input protection chain



5V => 3.3V

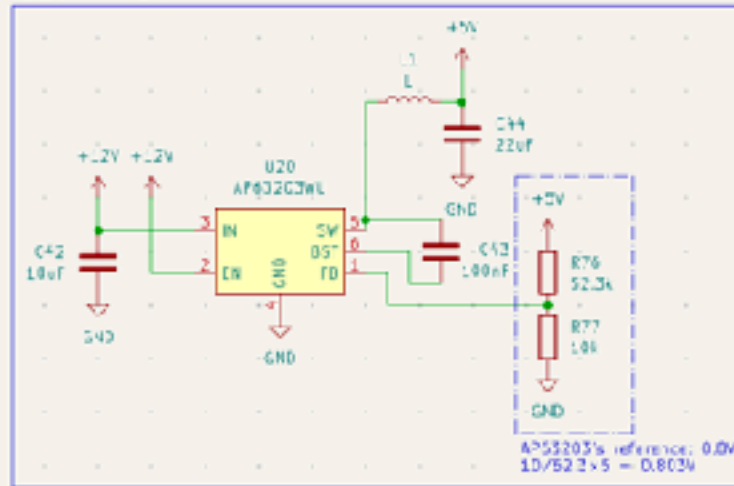
LDO



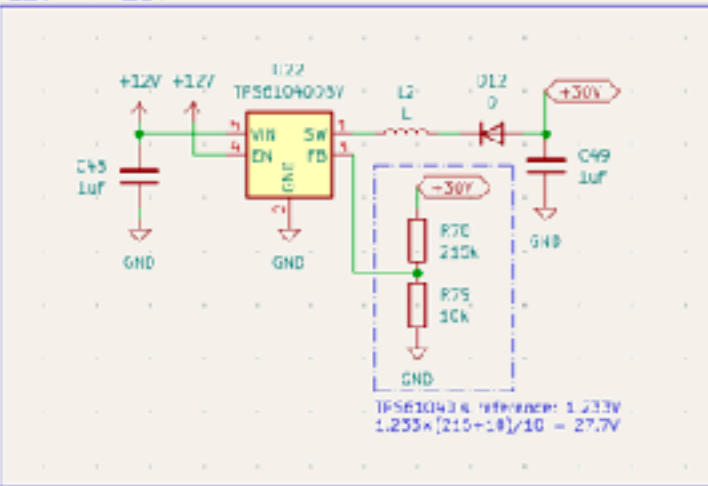
Analog rail



12V => 5V



12V => 28V



Grounds



Same voltage, separate copper
Noisy current return on GND
Measurement current return on analog GND

Buck

Boost (inductors + diodes)

mux can't pass a signal above its own supply, and the stack can't supply itself.

Not needed for diff-amp version